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Chapter 1:

About the Project

Inventory Management System - Java + MySQL

This is a simple Inventory Management System designed for grocery stores, built as a Software Design and Architecture semester project by a team of four. It follows proper Object-Oriented Programming, UML modeling, and design principles to support real-world maintainability and scalability.

Key Modules

The system is divided into 7 functional modules, each mapped to a clear use case and independently implementable:

Login / Logout

Role-based authentication (Admin, Employee)

Session management

Account Management

Admin can create, edit, or delete employee accounts

Enforces role-based access

Add Item to Inventory

Add new grocery items with attributes like name, quantity, category, price

Delete Item from Inventory

Delete outdated or discontinued items by Admin

Search Item

Search inventory by name, category, or ID

View Inventory

Display complete list of items with filtering options

Generate Reports

Create summary reports based on stock, activity, and item trends

⚙️ Tech Stack

Layer Technology

Language Java (JDK 17+)

Database MySQL

UI Java Swing

Database Access

JDBC

Design Tools Draw.io / Lucidchart for UML

Design Methodology

Object-Oriented Programming (O

P)

We used OOP to ensure modularity, reusability, and clean separation of concerns:

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Encapsulation: Each module has its own data and behavior

Abstraction: Users interact with high-level operations

Inheritance: Roles (e.g., Admin, Employee) can extend base

User

Polymorphism: Future enhancements can allow custom UI actions per role

Software Design Principles & Patterns

Single Responsibility Principle (SRP): Each class has a well-defined job

GRASP Patterns:

Controller: Each module has its own controller class

Creator: DAOs are responsible for creating model objects

Information Expert: Logic resides in the class that has the data

MVC Architecture:

Model: User, Item, Inventory

View: LoginUI, InventoryUI, etc.

Controller: LoginController, InventoryController

UML Diagrams (Created)

For each use case (module), the following were developed:

General Use Case Diagram (with <<include>> and <<extend>> as needed)

Fully Dressed Use Cases (structured table format)

System Sequence Diagrams (SSD)

Communication Diagrams (based on GRASP)

Class Diagrams (showing structure and relationships)

Roles & Access Control

Role Capabilities

Admin Full access: account mgmt, inventory CRUD, reports

Employee Limited access: login, view/search inventory, add items

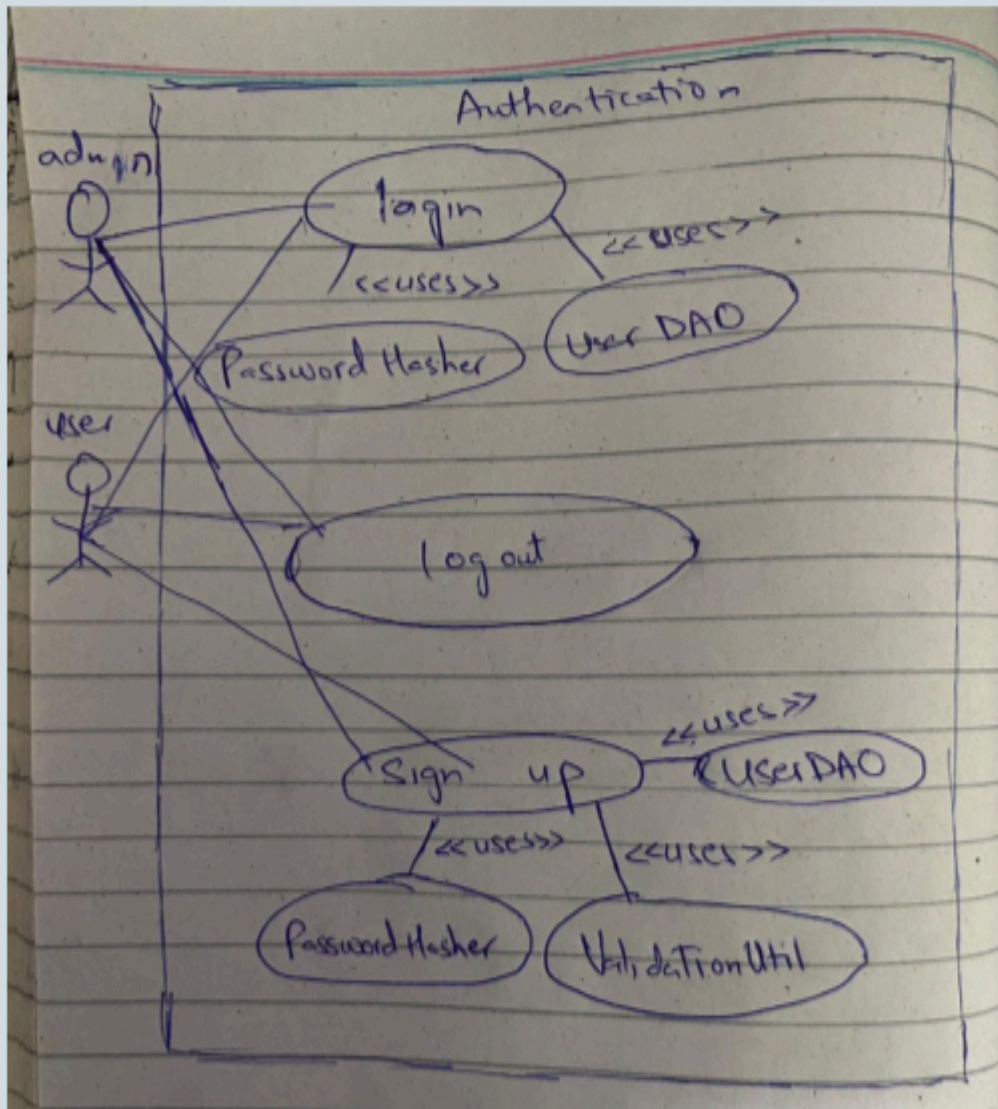
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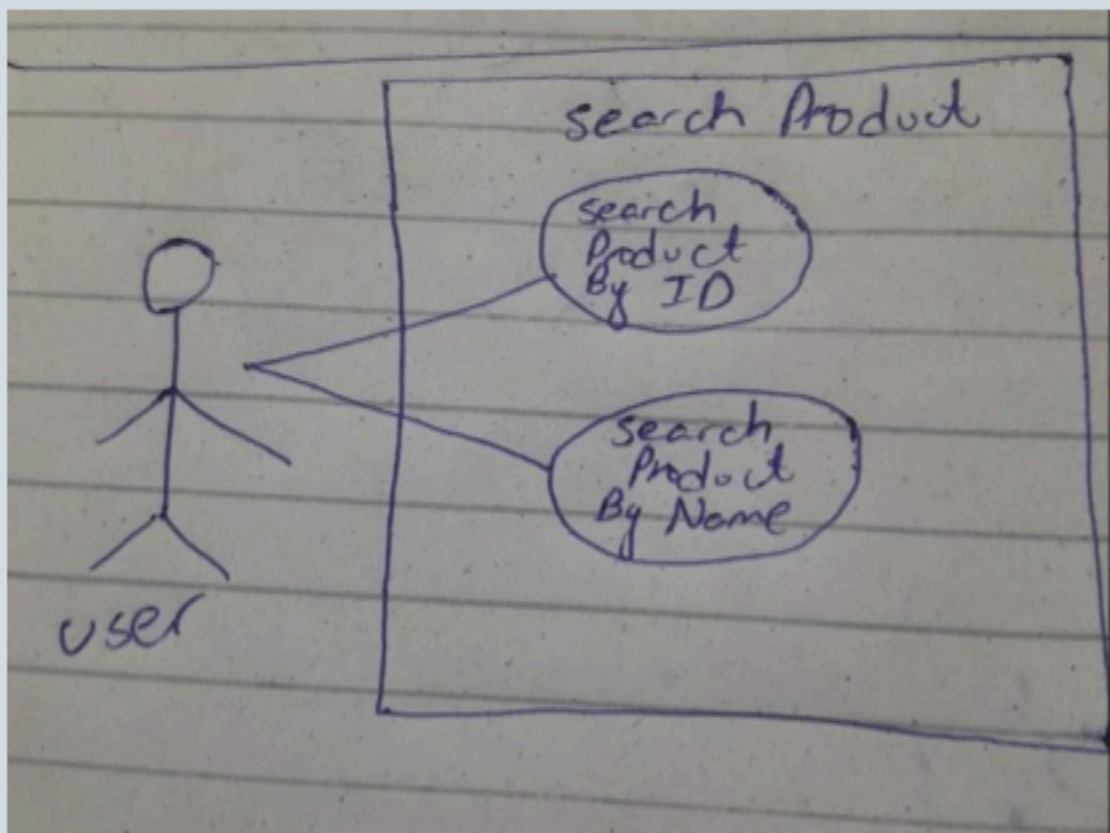
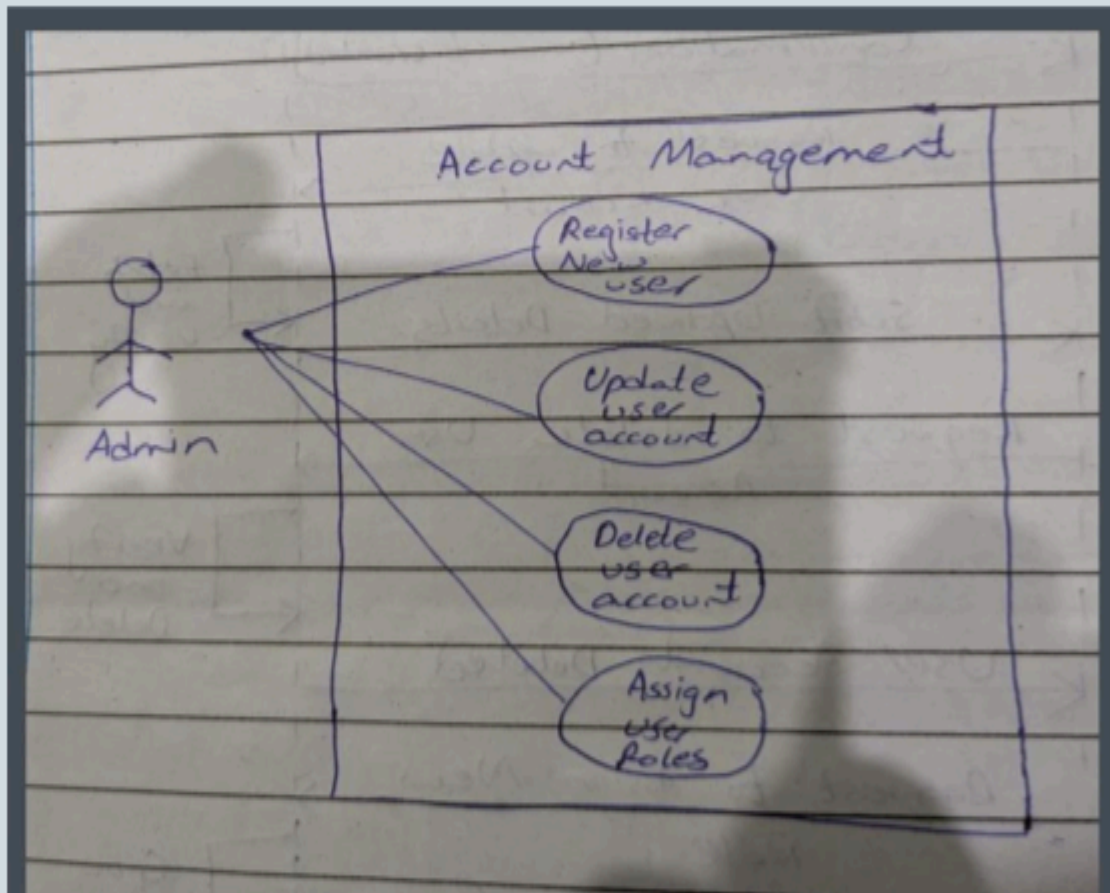
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Chapter 2

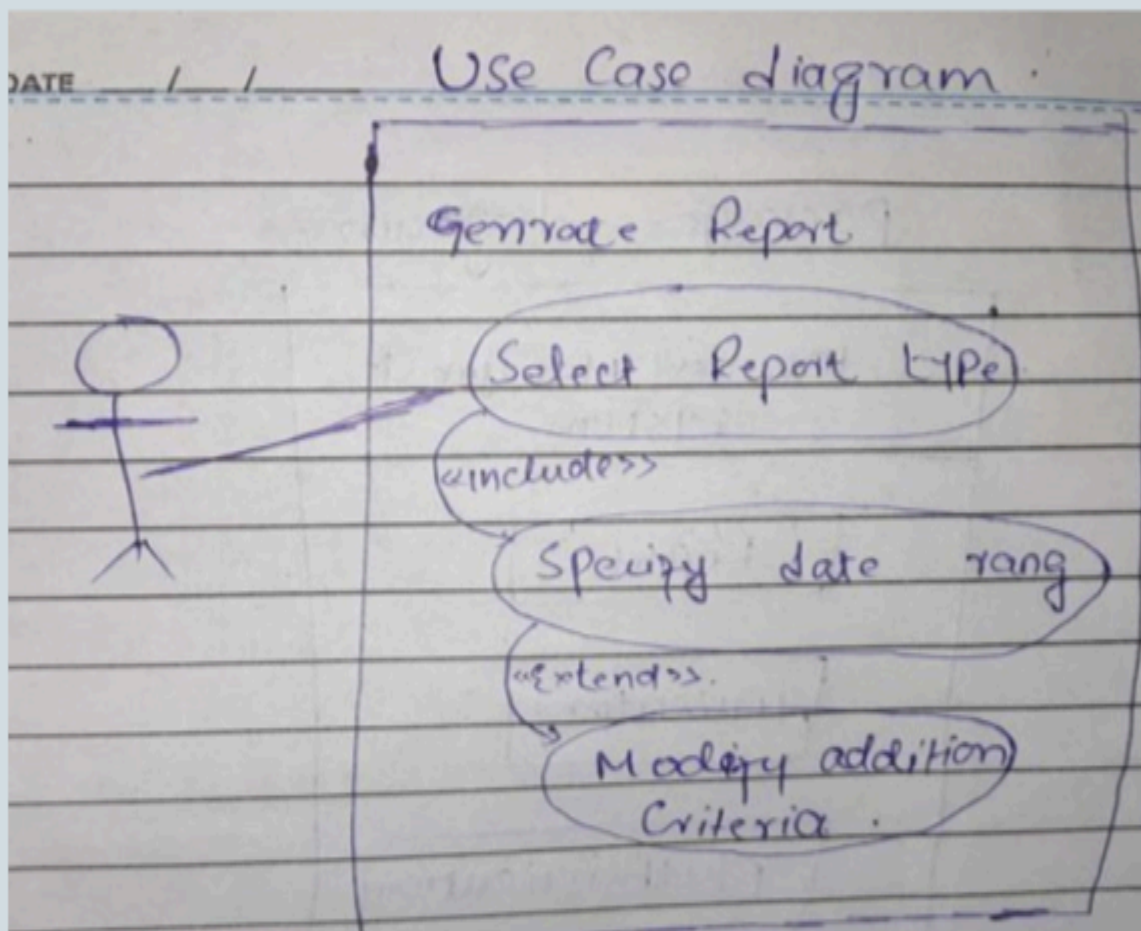
Use Case Diagram:

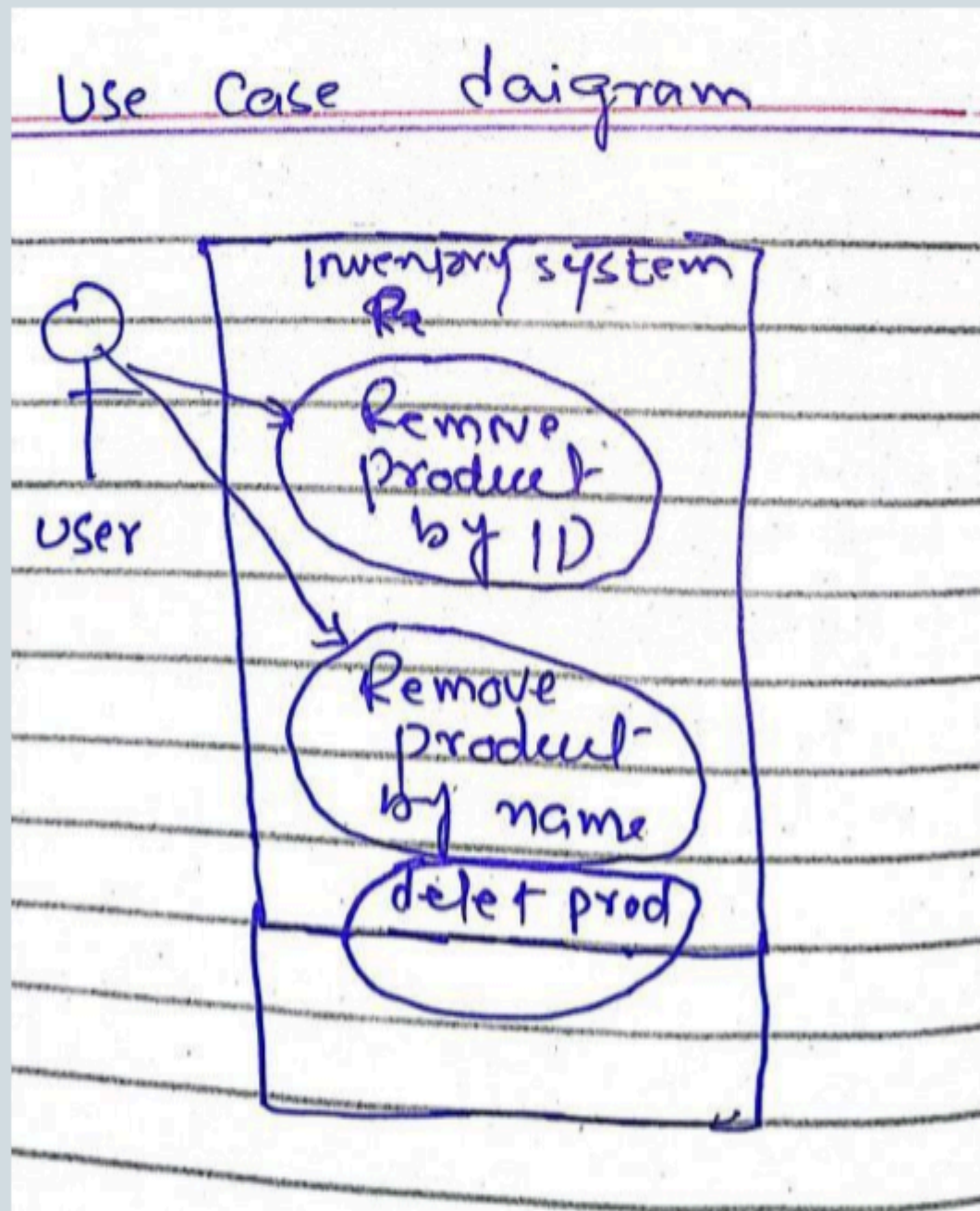
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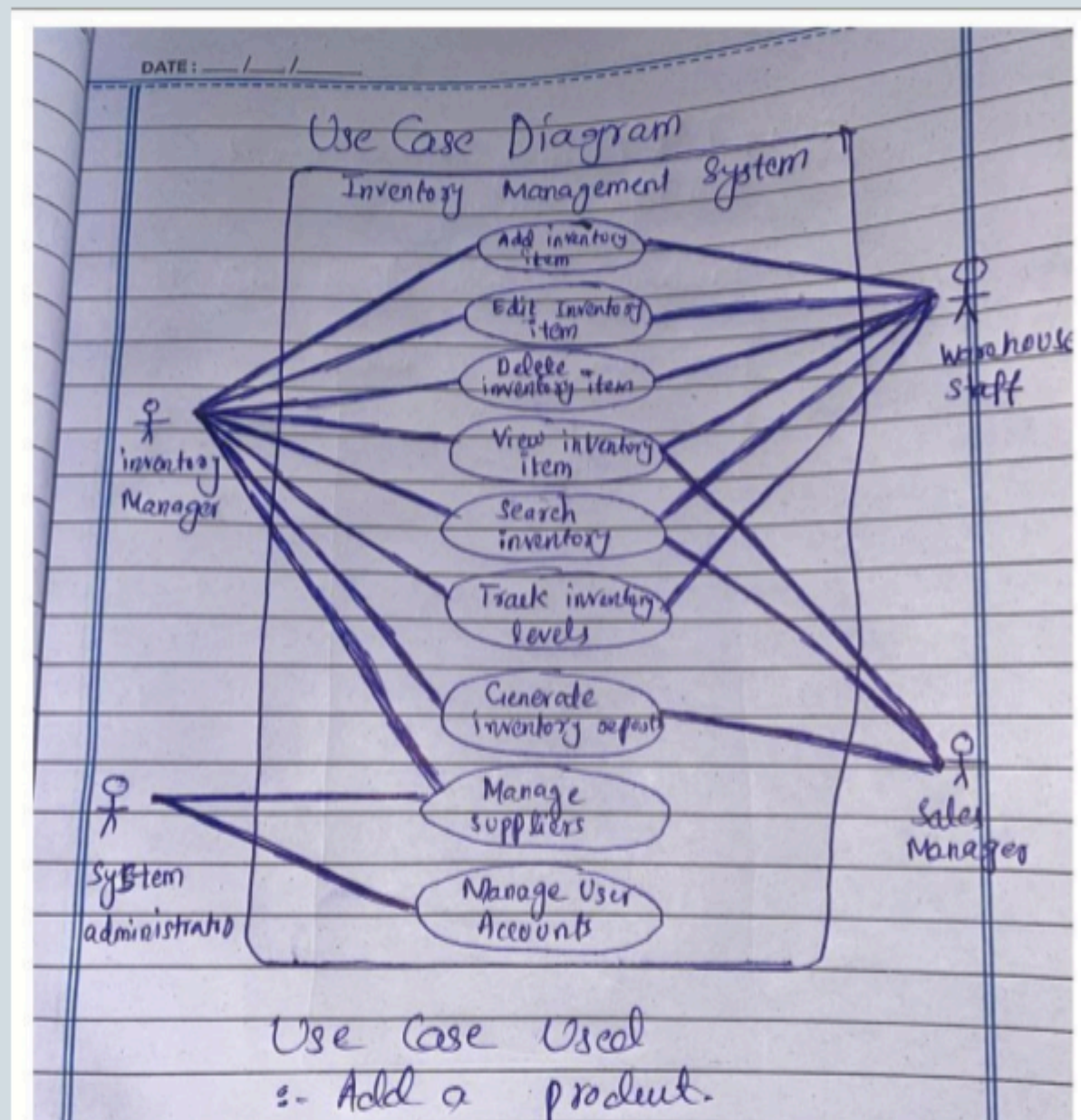


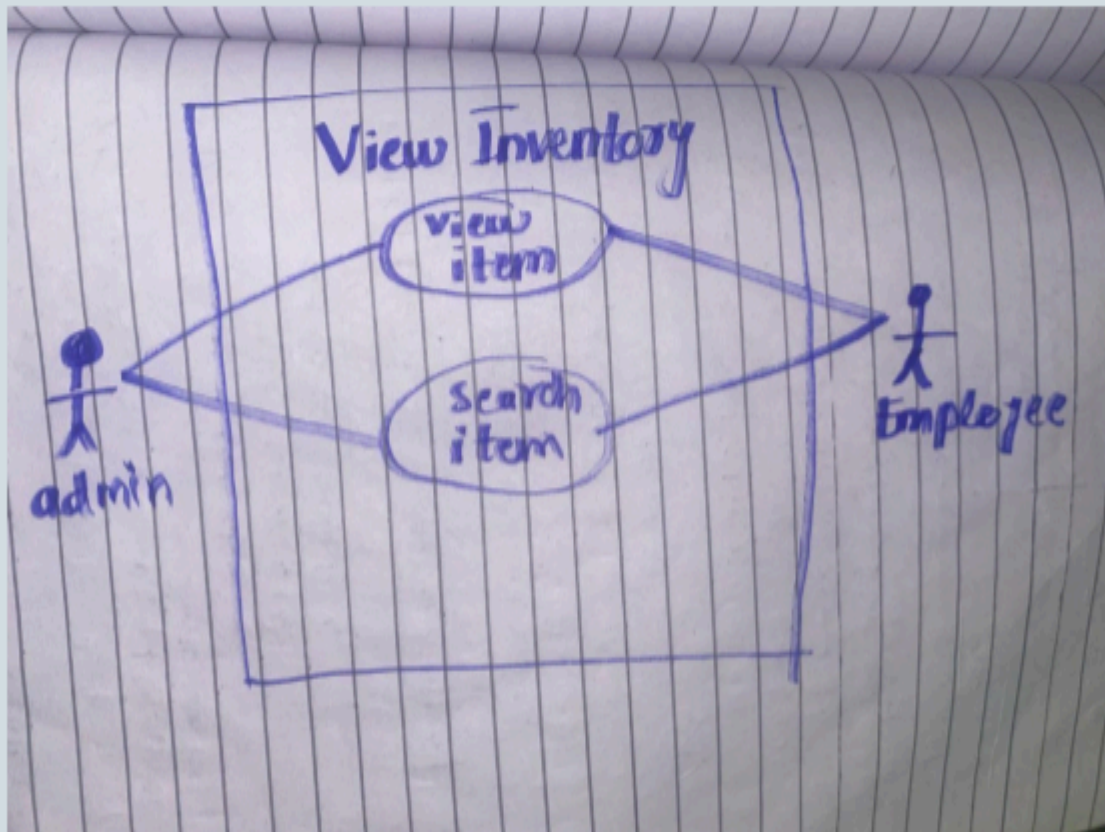
SP23 BSE 076 - Naseem Shahid





FA22 BSE 074 - Amanullah Khan





Chapter 3

Full Dressed Usecase:

FA22 BSE 059 - Syed Shah Mansoor Trimzi

Use Case Name: Authentication

Scope: Inventory Management System for Grocery Store

Level: User Goal

Primary Actor: Admin, Employee

Stakeholders and Interests:

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- Admin/Employee: Need secure access to system functions.
- System: Must verify user identity and prevent unauthorized access.

Preconditions:

- System is running and connected to the database.
- For login: user account exists and is approved.
- For sign-up: username must be unique.

Postconditions:

- On login: user is authenticated, and session starts.
- On logout: session ends securely.
- On sign-up: account is created with pending status.

Main Success Scenario (Basic Flow):

Login:

1. User accesses the login screen.
2. Enters username and password.
3. System fetches user from database using `UserDAO`.
4. Confirms status is “approved”.
5. Verifies password using `PasswordHasher`.
6. Identifies user role and grants access.
7. User uses the system.
8. User logs out.
9. System ends the session.

Sign Up:

1. User opens sign-up form.
2. Enters username, password, email, and role.
3. System validates input and checks username availability.
4. Hashes password and creates user with pending status.
5. Saves user to database using `UserDAO`.

Extensions (Alternative Flows):

Login – Invalid Credentials:

- If username or password is incorrect, system displays error and prompts retry.

Login – Account Not Approved:

- If user exists but is not approved, access is denied.

Sign Up – Username Exists:

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- If username already exists, system shows an error and requests a new one.

Sign Up – Invalid Input:

- If input is invalid or incomplete, system prompts correction.

System Error (Both Flows):

- On connection failure or exceptions, system logs error and shows a failure message.

Special Requirements:

- Passwords are stored using SHA-256 hashing (PasswordHasher).
- Only approved users can log in.
- UI should distinguish admin and employee roles after login.
- All user-related DB operations go through UserDao.
- Data is temporarily stored in User model objects.

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Primary Actor: Admin

Secondary Actors: System, Email Service

Preconditions:

Admin is logged in with sufficient privileges.

System is operational.

Main Success Scenario:

1. Admin selects an account management action (Register/Update/Delete/Assign Roles).

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2. System displays the relevant form based on the

action: Register New User: Name, email, role fields.

Update User: Editable fields (e.g., name, email).

Delete User: Confirmation dialog.

Assign Roles: Role dropdown list.

3. Admin enters required details and submits the request.

4. System validates inputs:

For registration: Checks email uniqueness.

For updates: Verifies user exists.

For deletion: Confirms no active dependencies.

For role assignment: Ensures admin has permission.

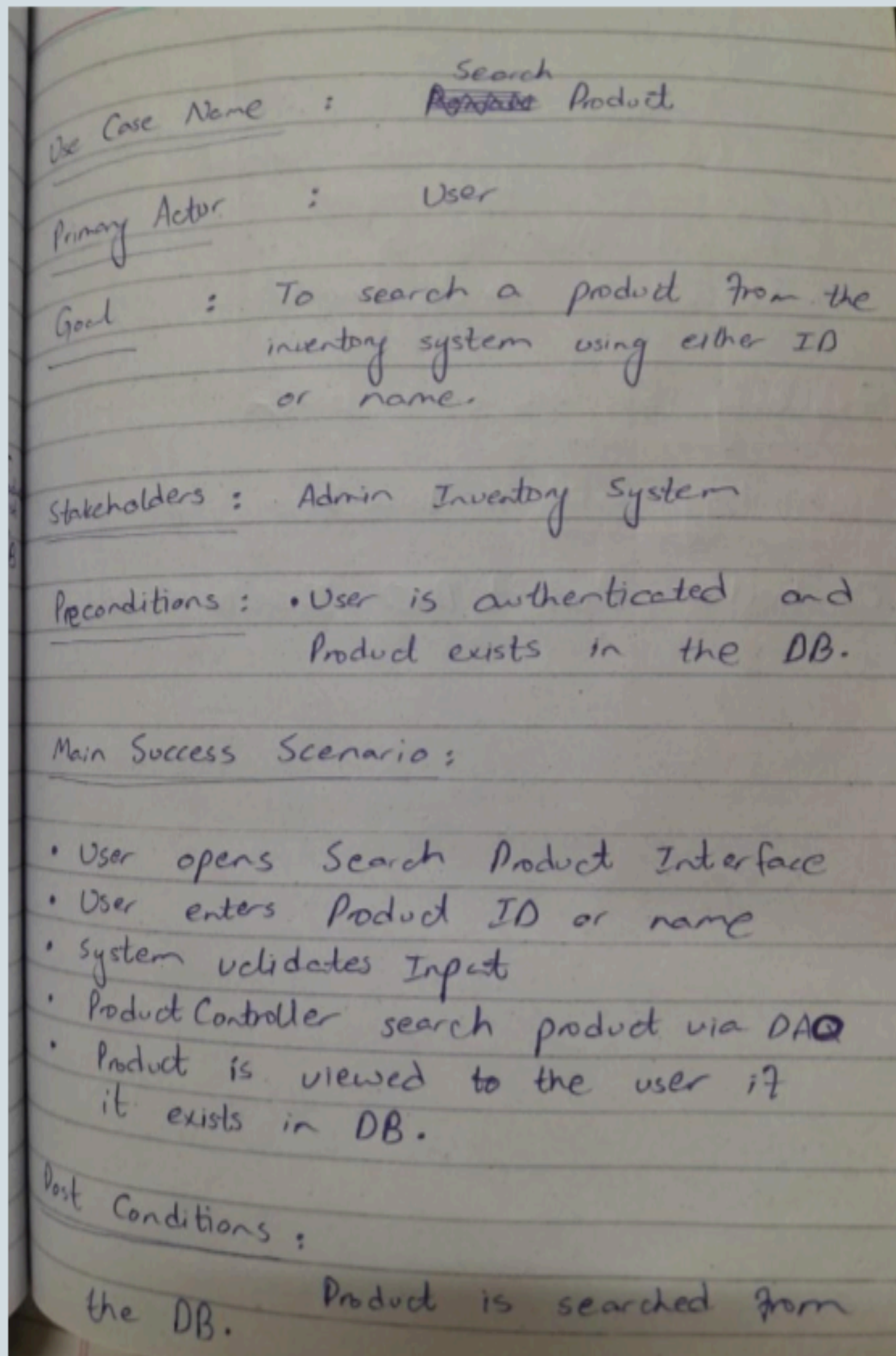
5. System processes the request:

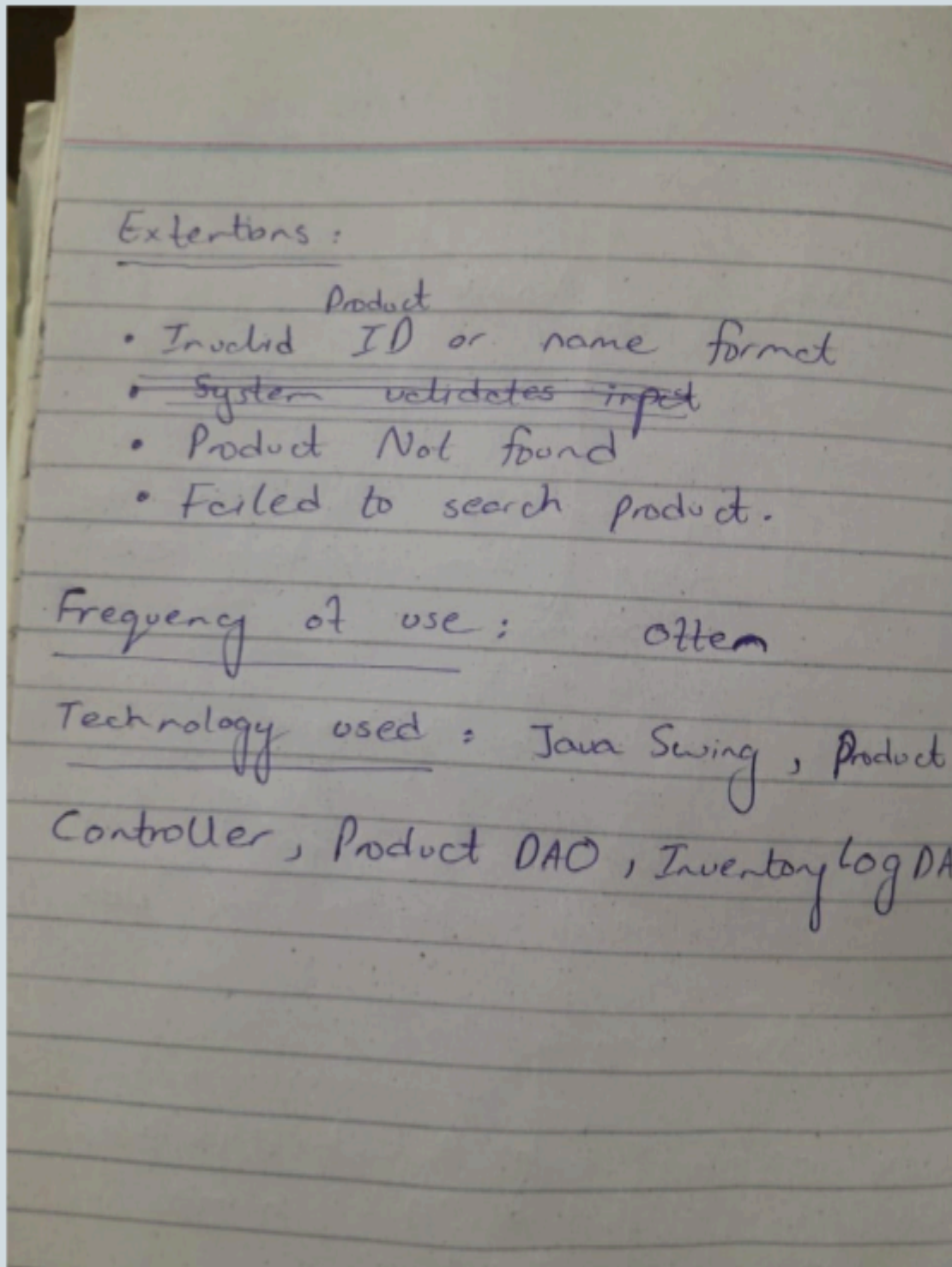
Creates/updates/deactivates the user account or updates roles. Sends notifications (e.g., welcome email for new users).

6. System confirms success to the Admin.

Postconditions:

- For registration: New user is added to the system.
- For updates: User details are modified.
- For deletion: User account is deactivated/removed.
- For role assignment: User permissions are updated.





Extentions:

- Invalid ^{Product} ID or name format
- ~~System validates input~~
- Product Not found
- Failed to search product.

Frequency of use: often

Technology used: Java Swing, Product Controller, Product DAO, InventoryLogDAO

Fully dressed usecase.

Use Case #: Generate Report

Description: Admin generate inventory and Sale Report.

Preconditions: Admin must be login
input Credentials.

Post Conditions: Report are generated and saved.

Main Success Scenario: Admin select "Generate Report".

- 1) System prompt for Report type and date Range
- 2) Admin Submit the Request.
- 3) system fetch data.
- 4) system generate and display/saves Report

Extensions: invalid input → show Error
No data available → show Message.

Fully dressed usecase diagram

Usecase names Remove product
Goal : To Remove product
 From inventory system
 by ID or name.

Stakeholders: User, Business admin
 Inventory System.

Preconditions -> product exist
 in the system
 -> User is authenticated

Post conditions -> product is deleted
 From data base
 -> action is logged in
 in inventory log

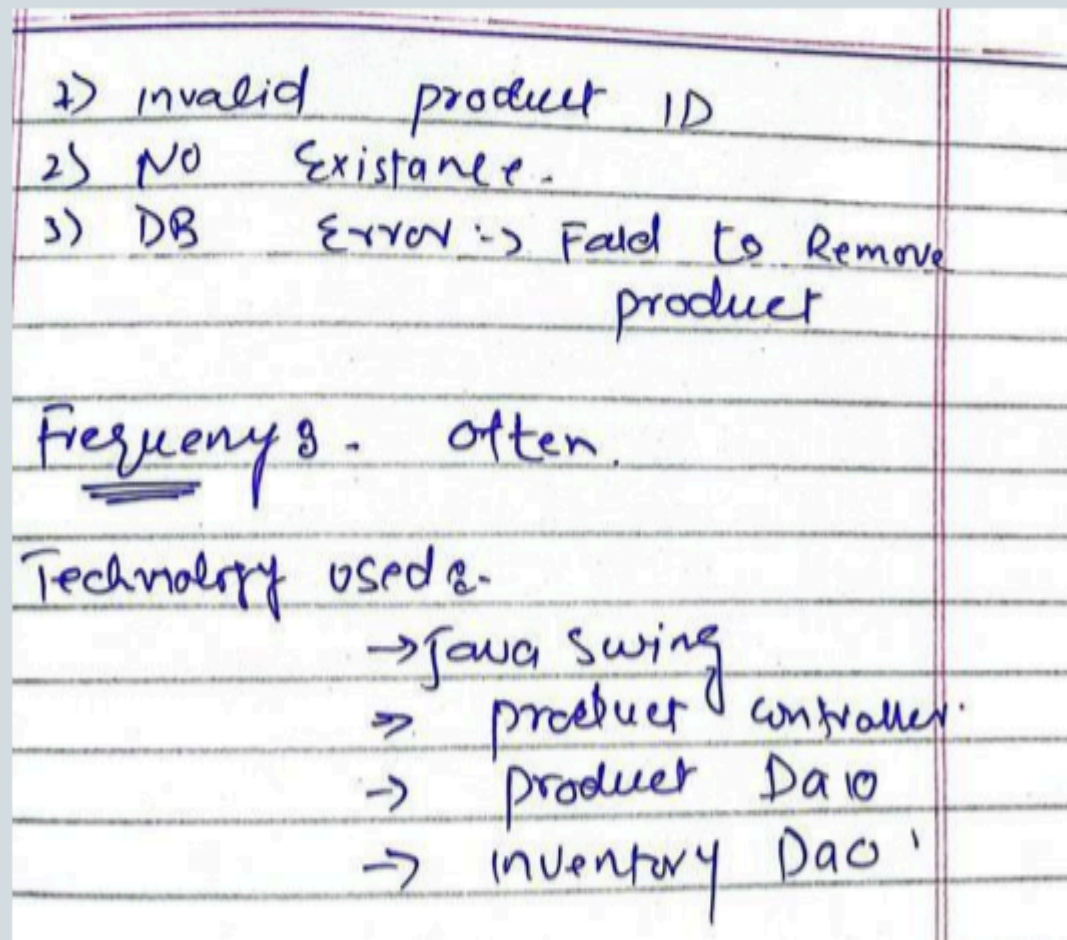
main Success Scenario -

- 1) user open Remove product interface
- 2) User Enter product ID, Name
- 3) System validate input
- 4) product controller delete product
- 5) log entry is saved.
- 6) Conformat Show to user.

extension -

- 1) No input -> system show
 please provide ID, Name

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Fully Dressed Use Case: Add Item to

Inventory Use Case Name Add Item to

Inventory Primary Actor Admin or Employee

Stakeholders and Interests Admin and Employee want to maintain accurate stock levels. The organization wants reliable inventory data for operations, sales, and reordering.

Preconditions - User must be logged in

- User must have appropriate role

(Admin/Employee) Postconditions - New item is saved in the database

- Inventory is updated with the new item

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Main Success Scenario (Basic Flow)

The user selects "Add Item" from the dashboard.

The system displays the item input form (name, quantity, price, category, etc.).

The user fills in all the required fields.

The user clicks the "Add" or "Save" button.

The system validates the input.

The system saves the new item to the inventory database.

The system confirms the addition and updates the inventory list.

The user sees a success message.

Extensions (Alternate Flows)

5a. Input is invalid (e.g., missing name or negative quantity)

The system highlights the invalid fields and shows error messages.

The user corrects the input.

Returns to Step 4.

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6a. Database connection fails

The system displays an error message: "Failed to save item. Please try again later."

The system logs the error.

Use case ends with failure.

Special Requirements

Form must validate input before submission.

Quantity must be numeric and non-negative.

Item name should be unique (or warn about duplicates).

The system must log all item additions with timestamps and user IDs for audit.

Frequency of Use

Frequently used by Admin and Employees during restocking or item updates.

Technology and Data Variations

Inventory is stored in a MySQL database.

Item addition form is implemented in a Java Swing UI.

Backend interacts with the database using JDBC.

Fully dressed Use Case

View inventory

Use Case Name	View Inventory
Scope	GrocerFlow Inventory Management System.
level	User-goal
primary Actor	Inventory Manager (or any authorized user)
Stakeholders and Interests	• Inventory Manager:-Wants to view a list of all products with details such as quantity, price, and added date to make decisions. • Business Owner: Wants to ensure inventory data is up-to-date and visible for analysis. • System: Must retrieve and display data from database correctly and reliably.
Pre-conditions	• User must have access to the system (GUI application is running) • Products must exist in the database
post conditions	• The product list is displayed • Data is fetched from the database and shown in a

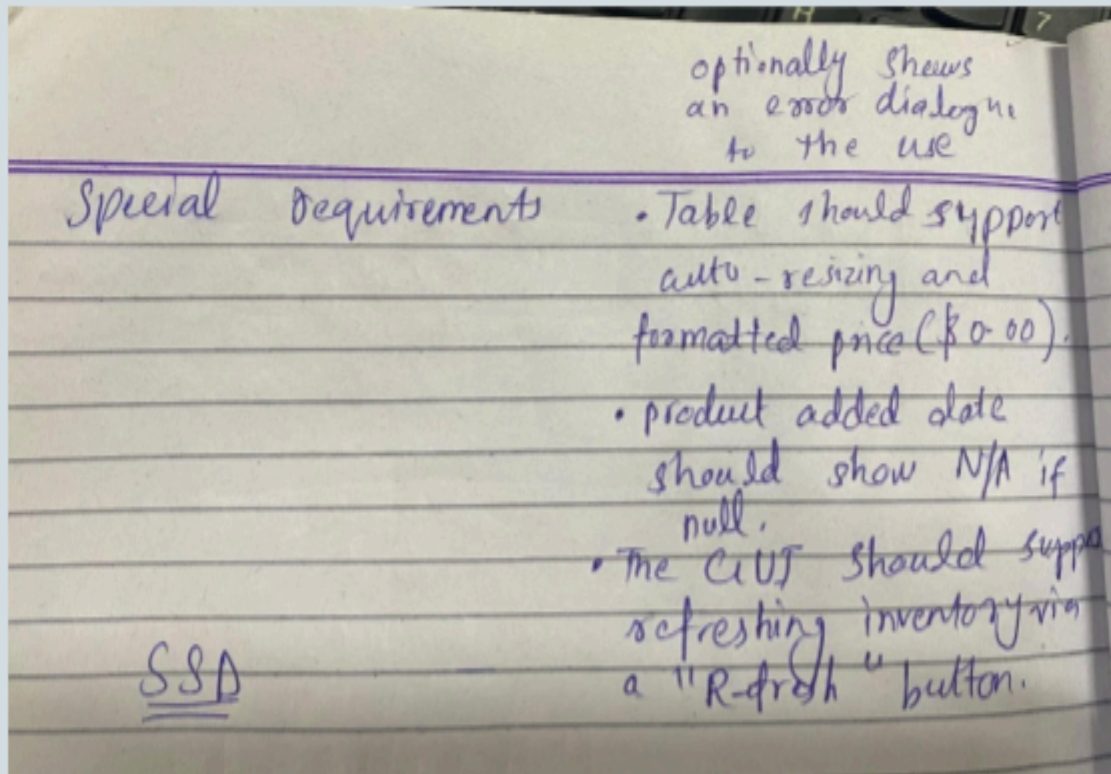
tabular format

Main success Scenario
(Basic flow)

- 1 - The user opens the "View Inventory" screen
- 2 - The system initializes the view Inventory view frame.
- 3 - Load Inventory () is called.
- 4 - Product Controller.get Allproducts() is called
- 5 - Controller calls ProductDAO.get AllProduct ().
- 6 - DAO retrieves products from database.
- 7 - Controller receives the list of product objects.
- 8 - View loops through the list and populates the J Table.
- 9 - User sees all products in the inventory displayed in the GUI.

Extensions (Alternative flows)

- 3a - No product found
- 3a1 - An empty table is displayed.
- 5a Database connection fails
- 5a1 - System logs the error

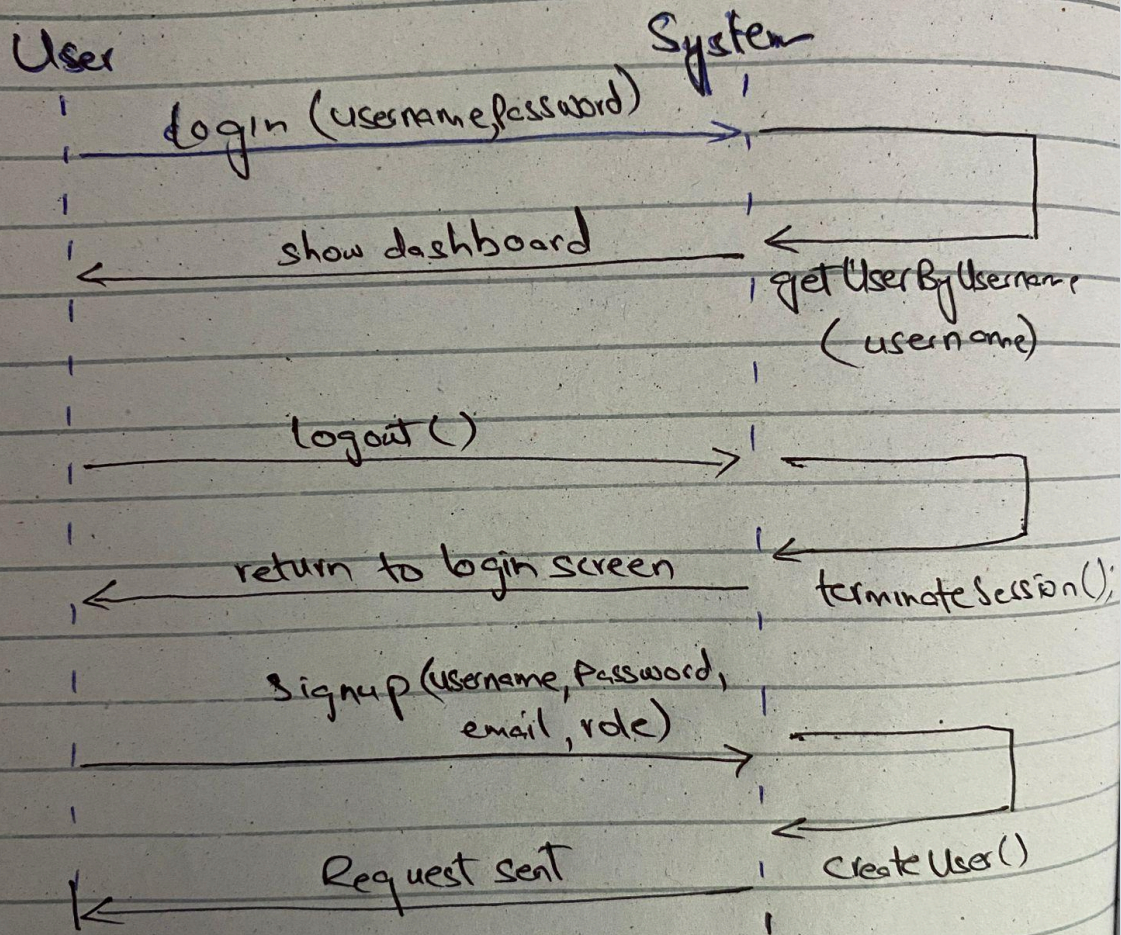


Chapter 4

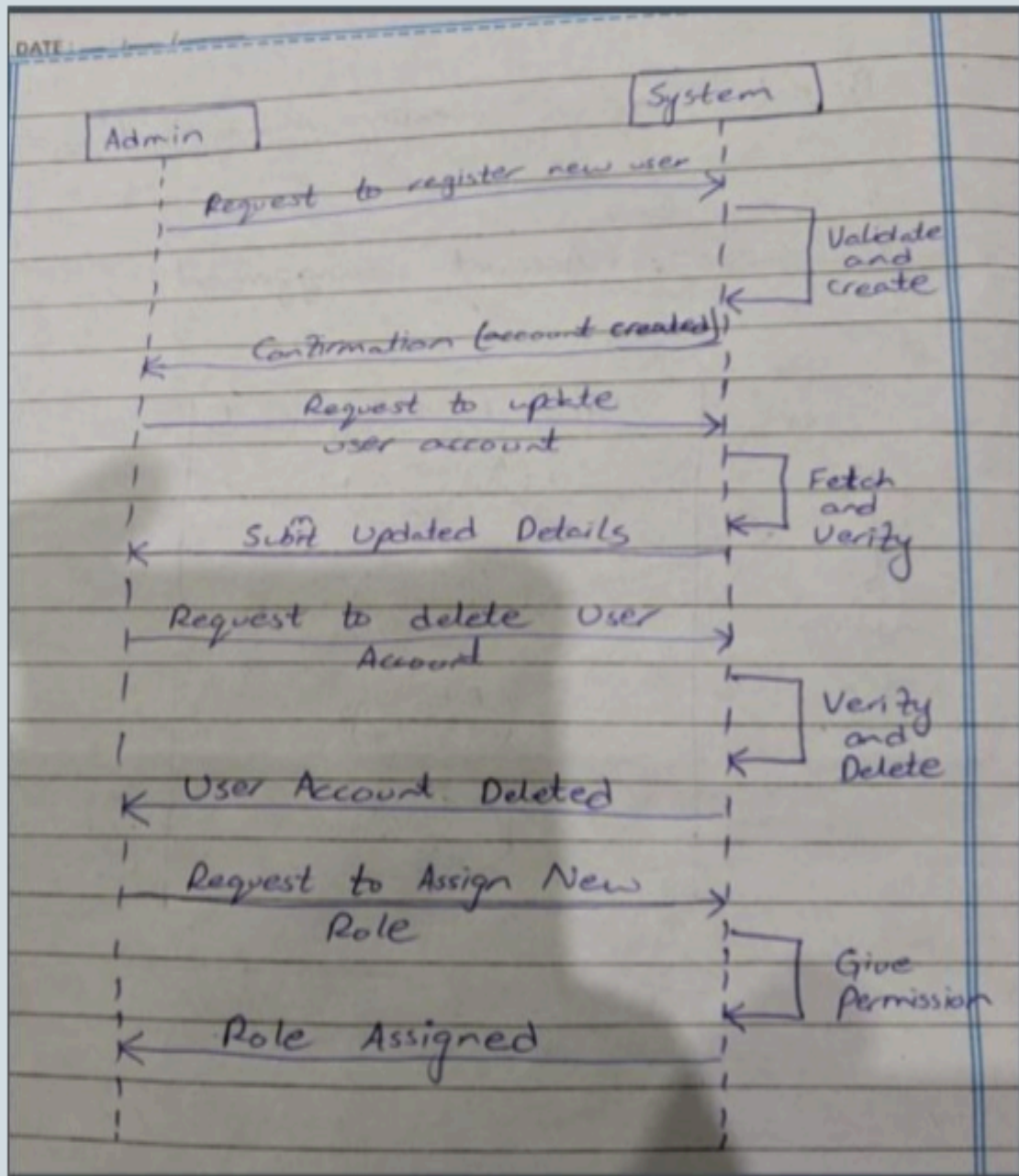
SSD

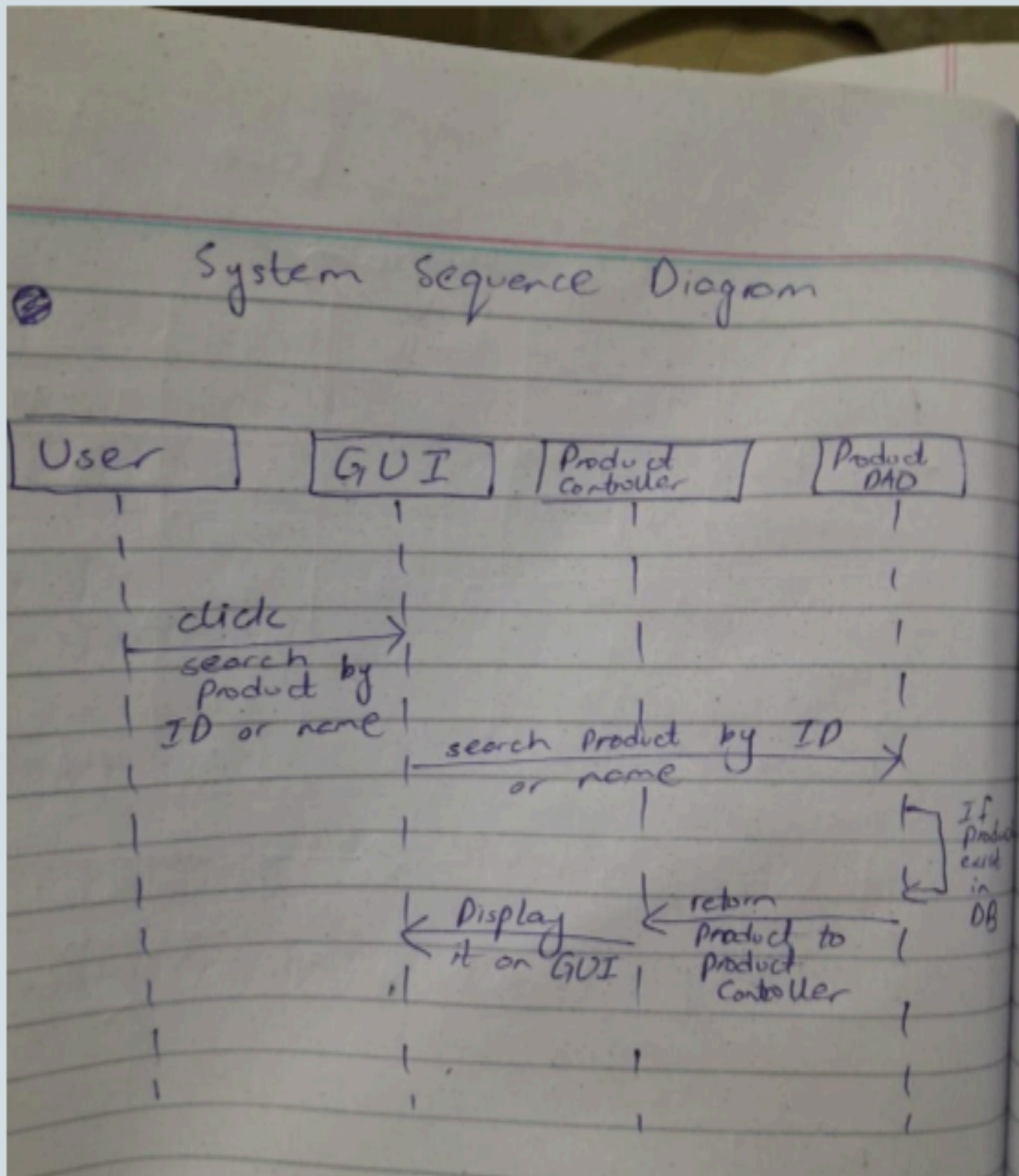
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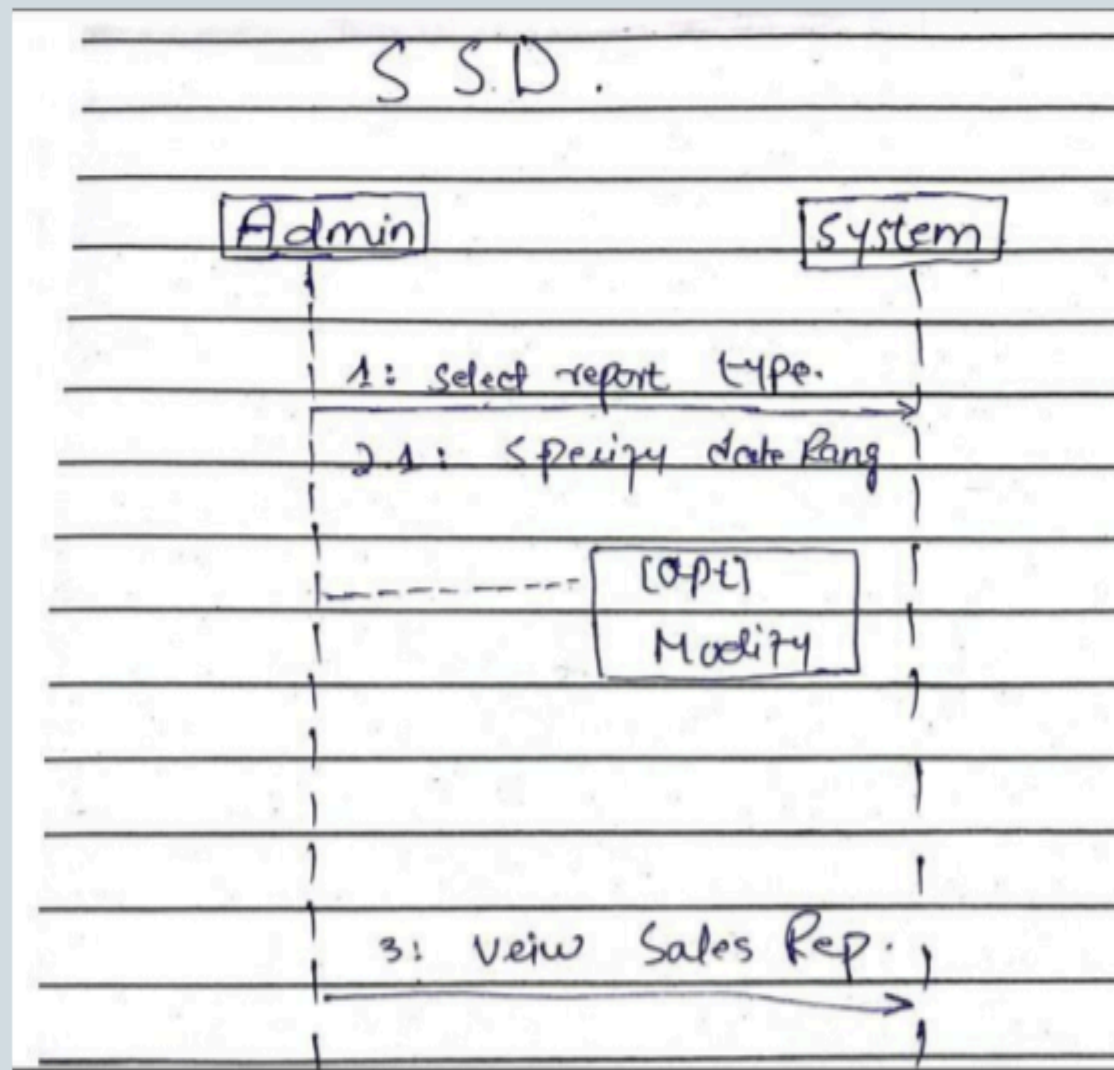
SSD for authentication

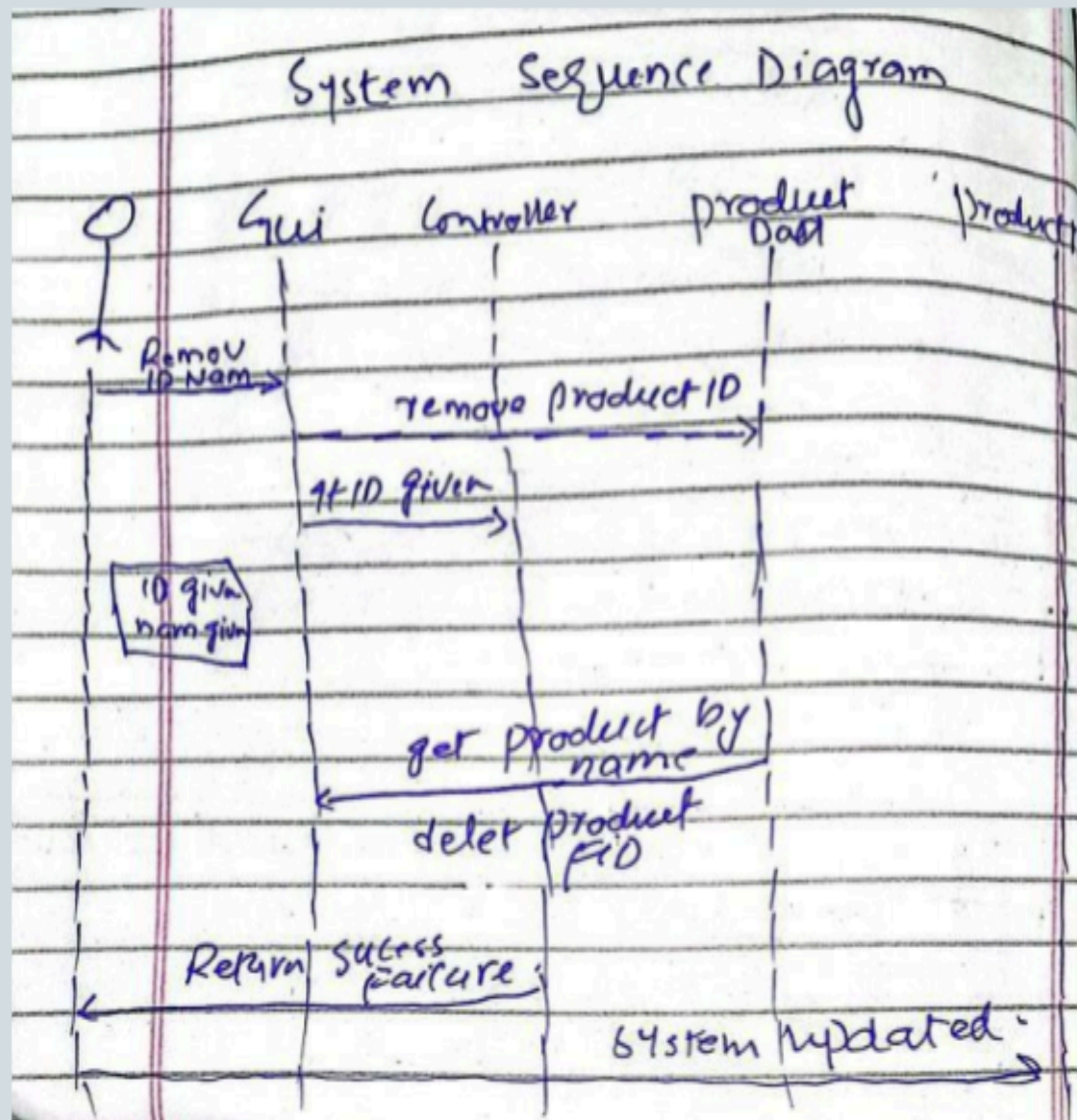


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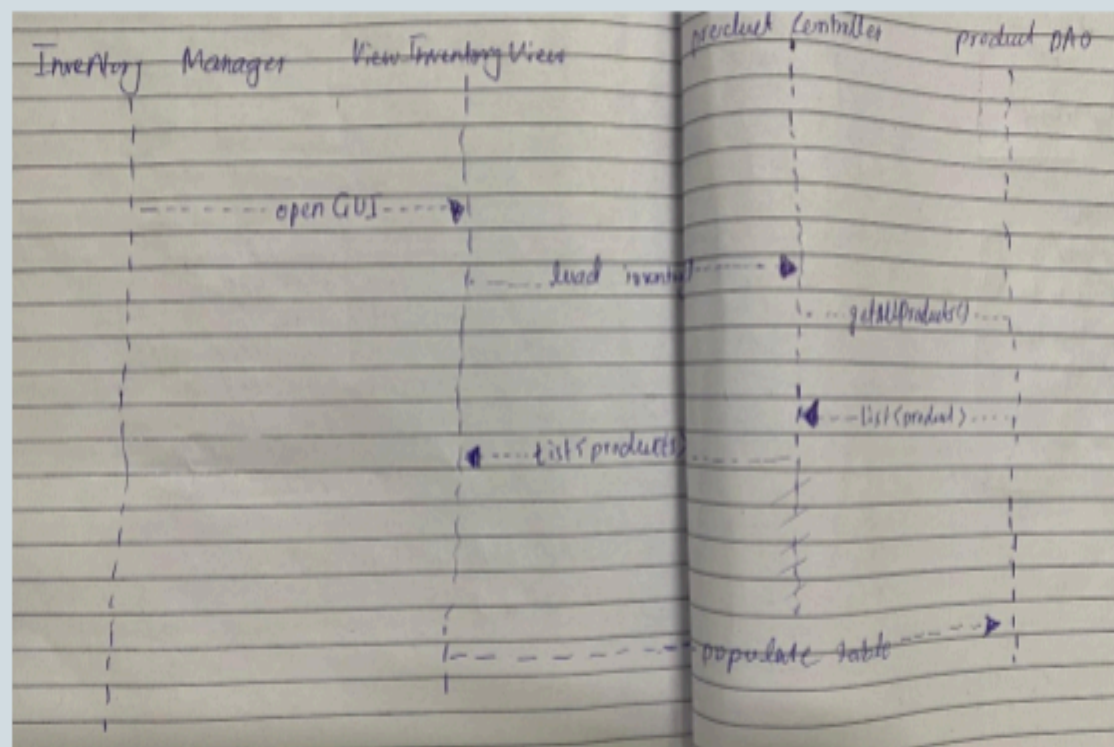
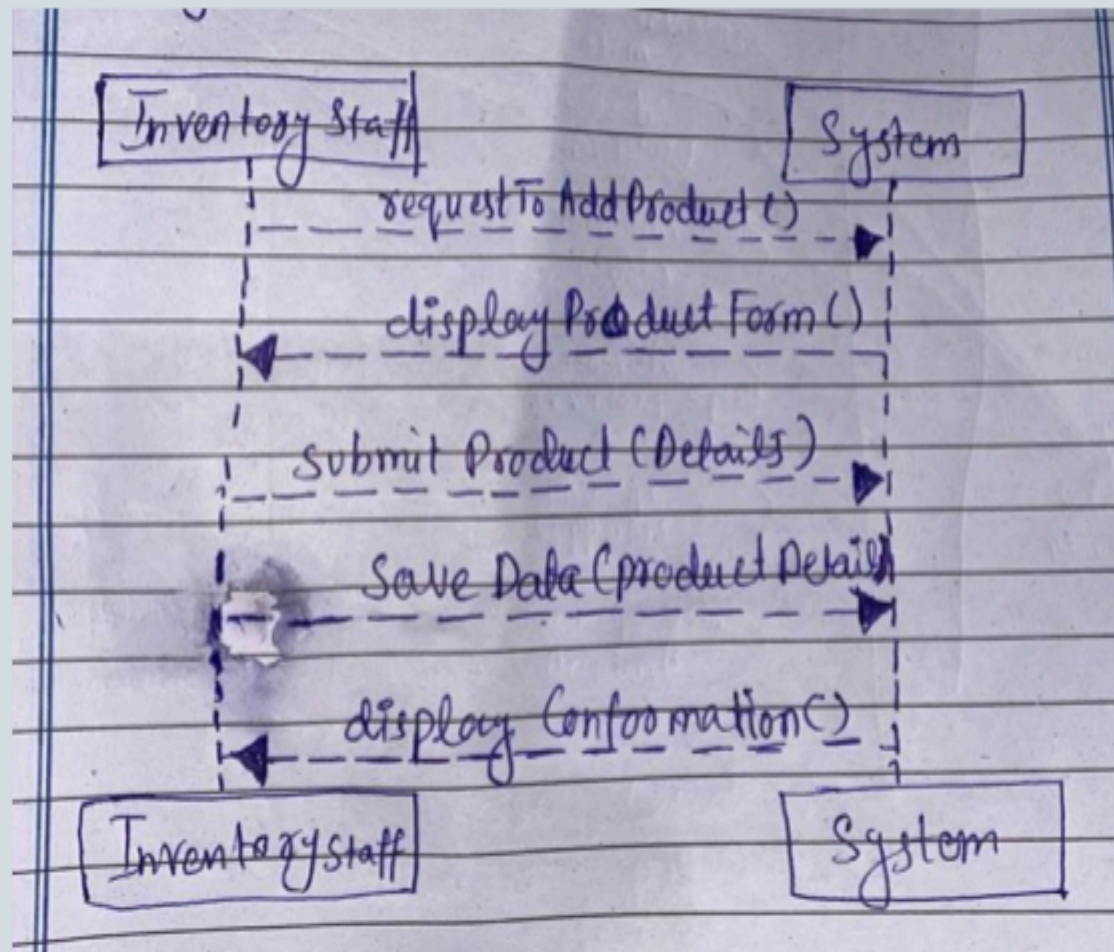








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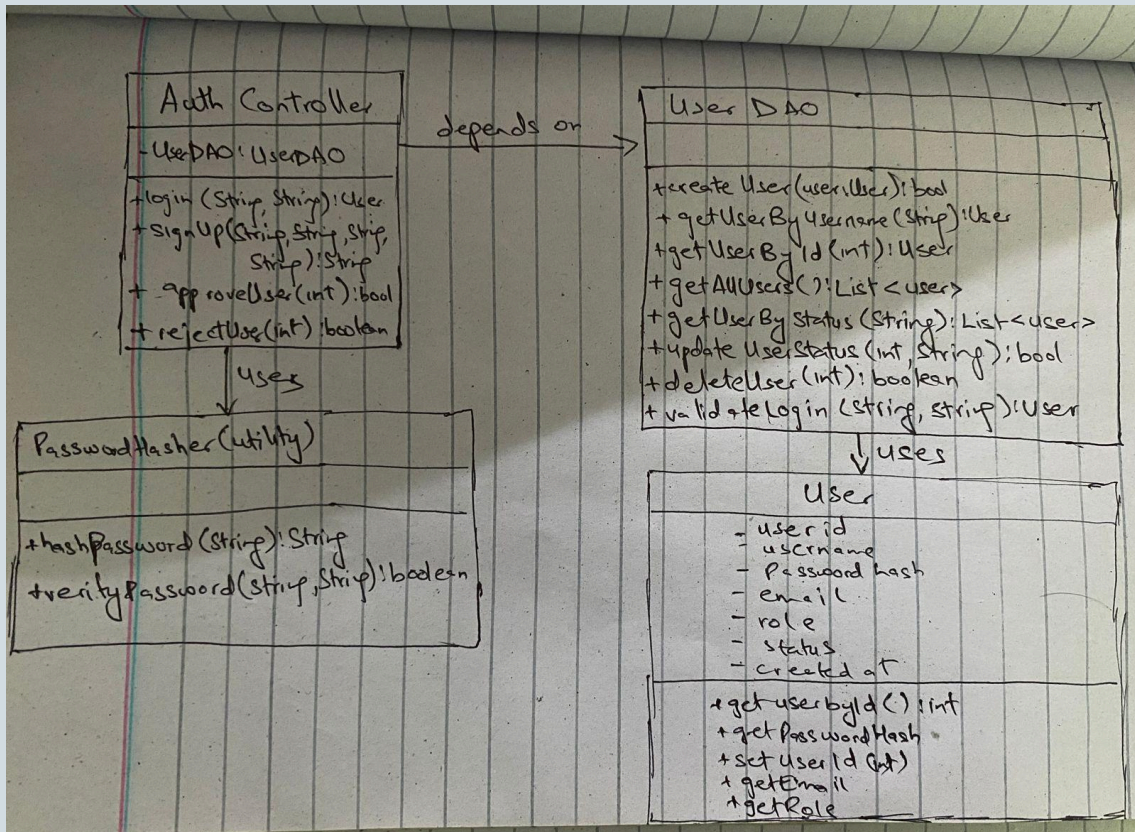
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Chapter 5

Class Diagram:

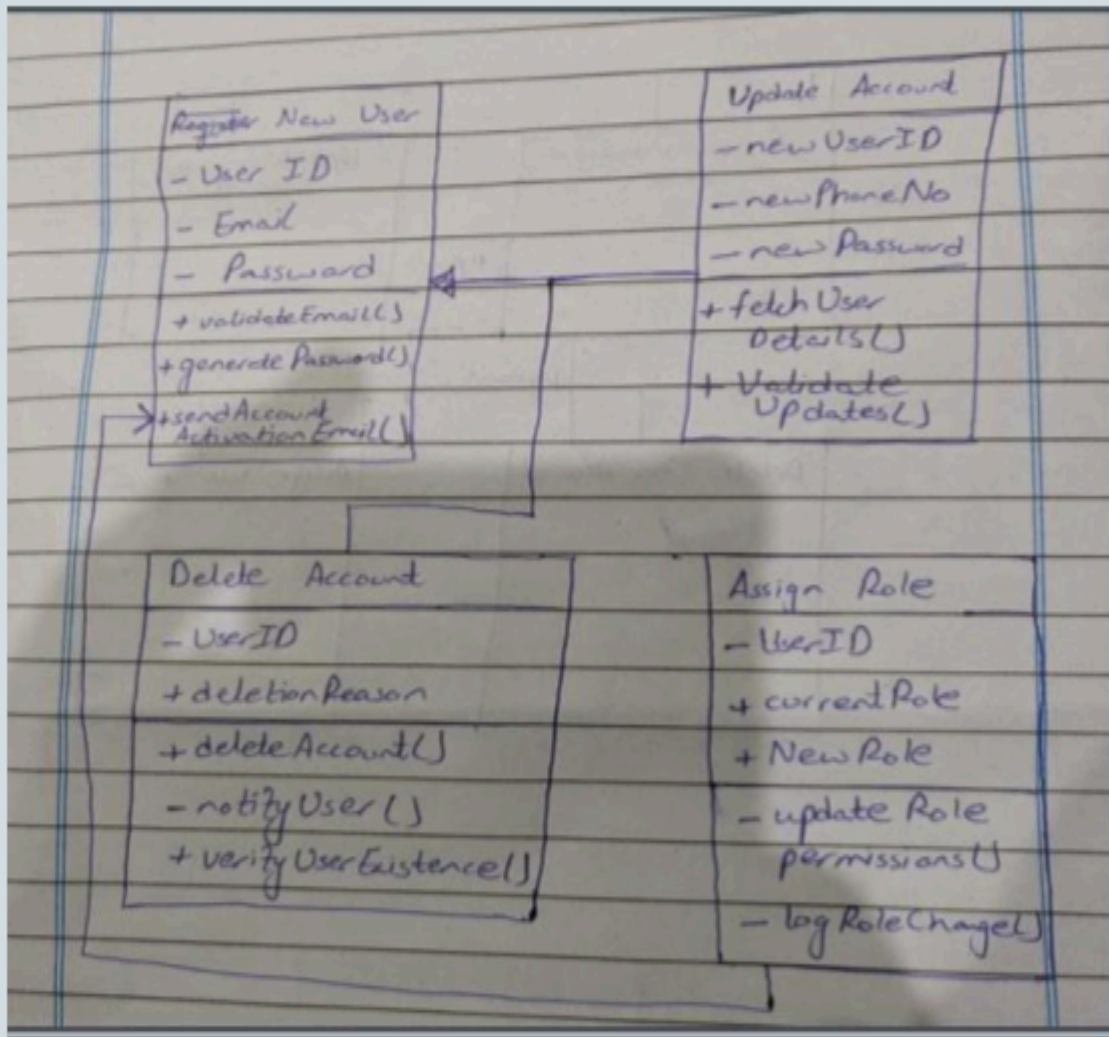
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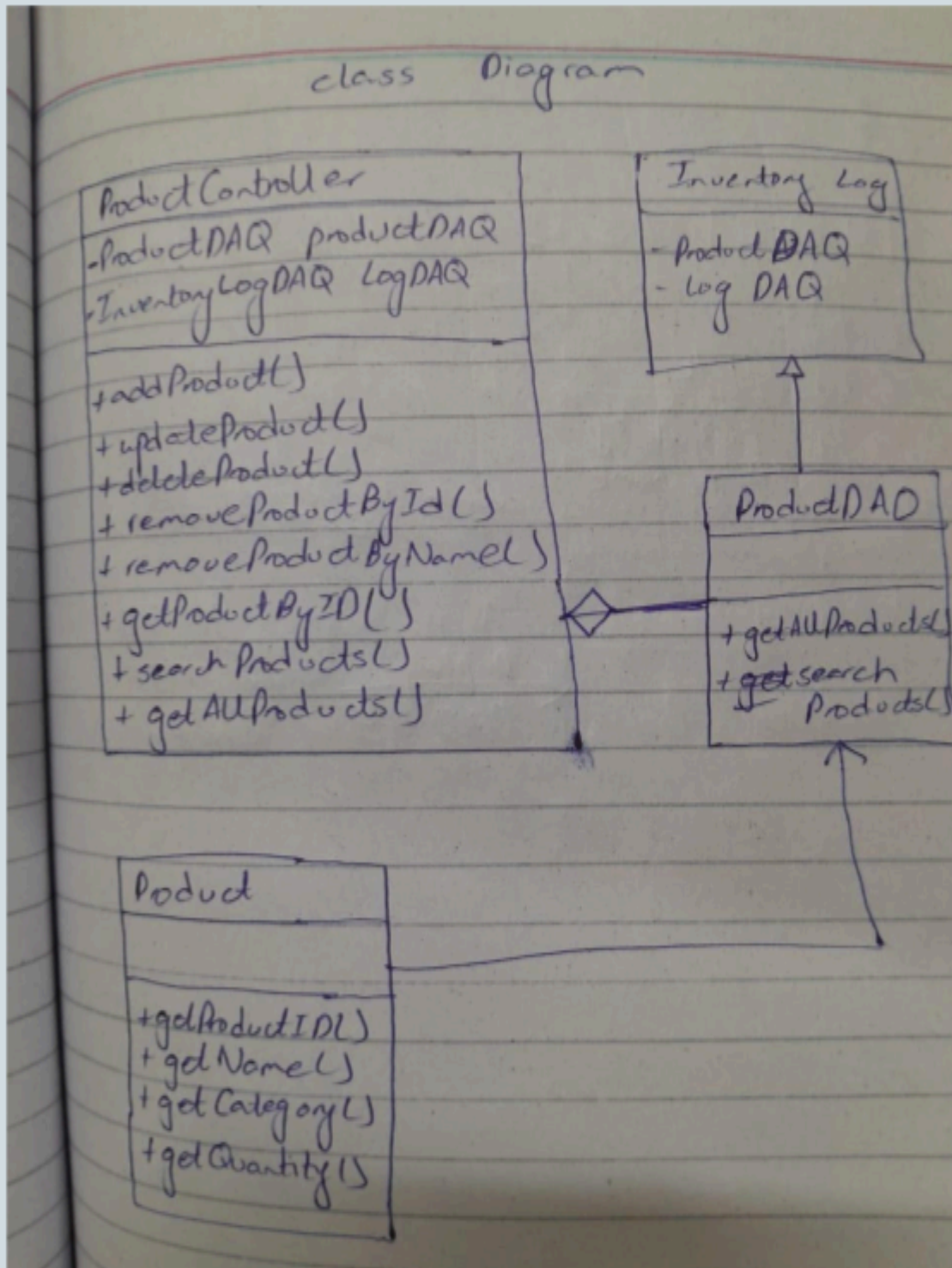
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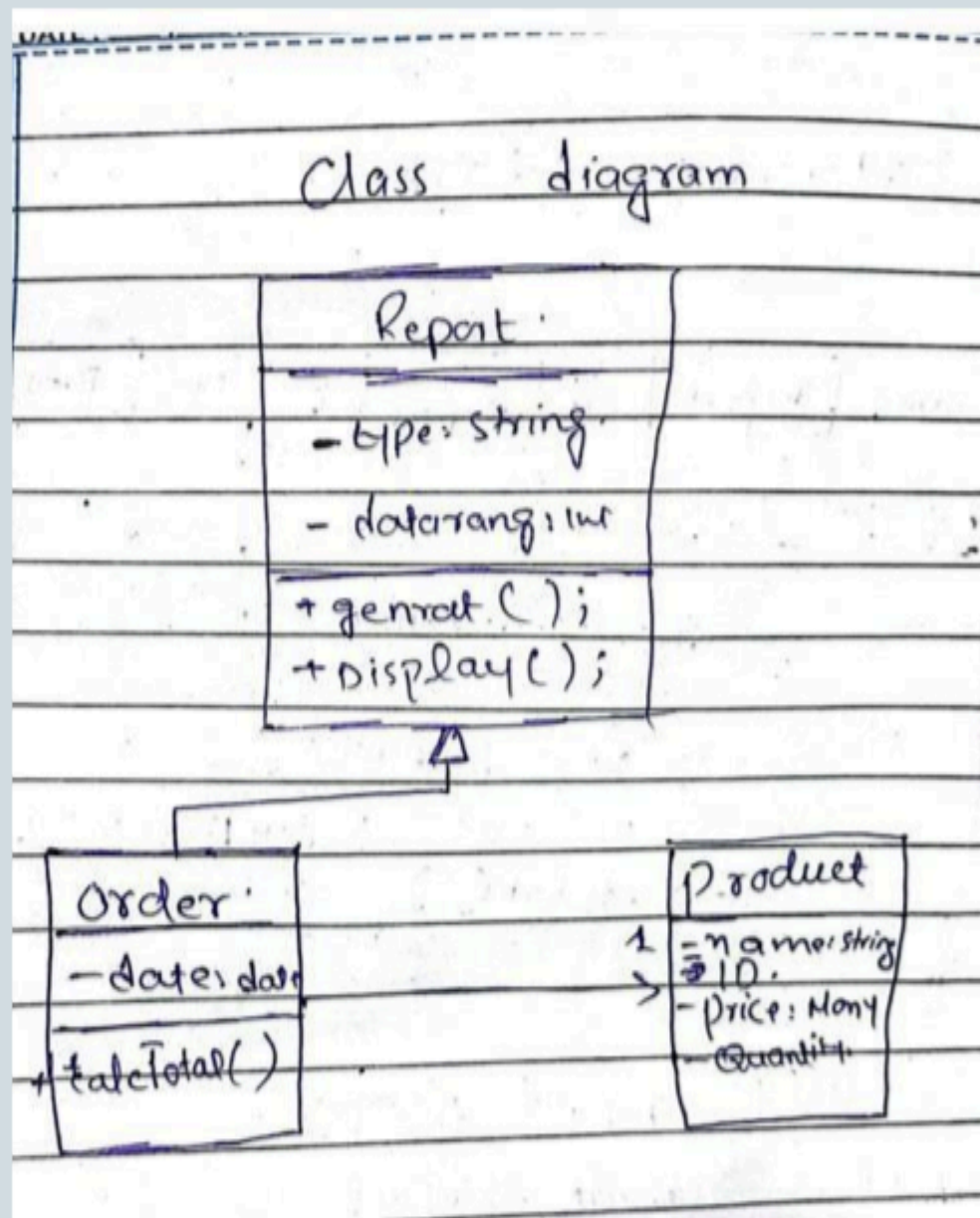


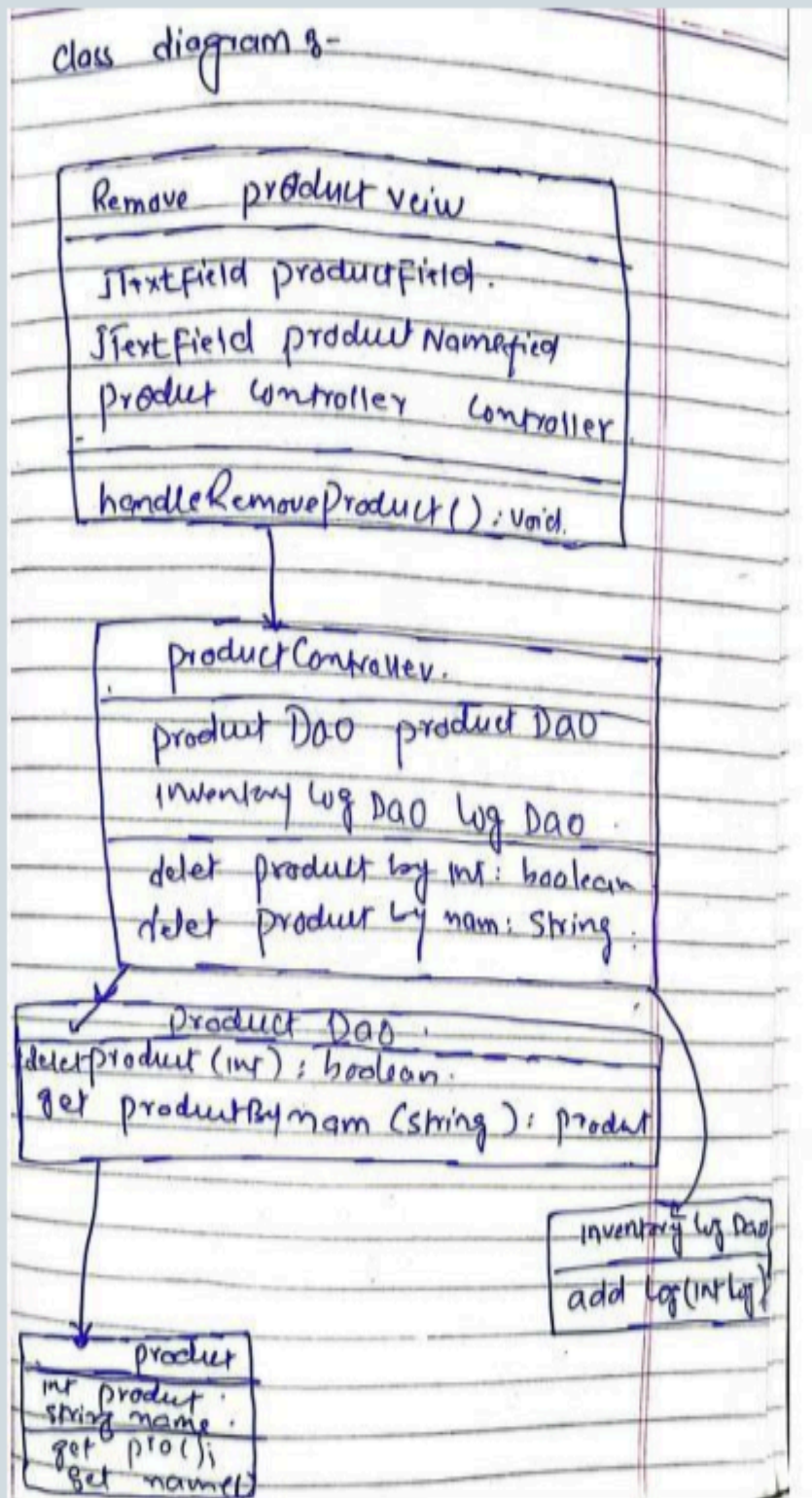
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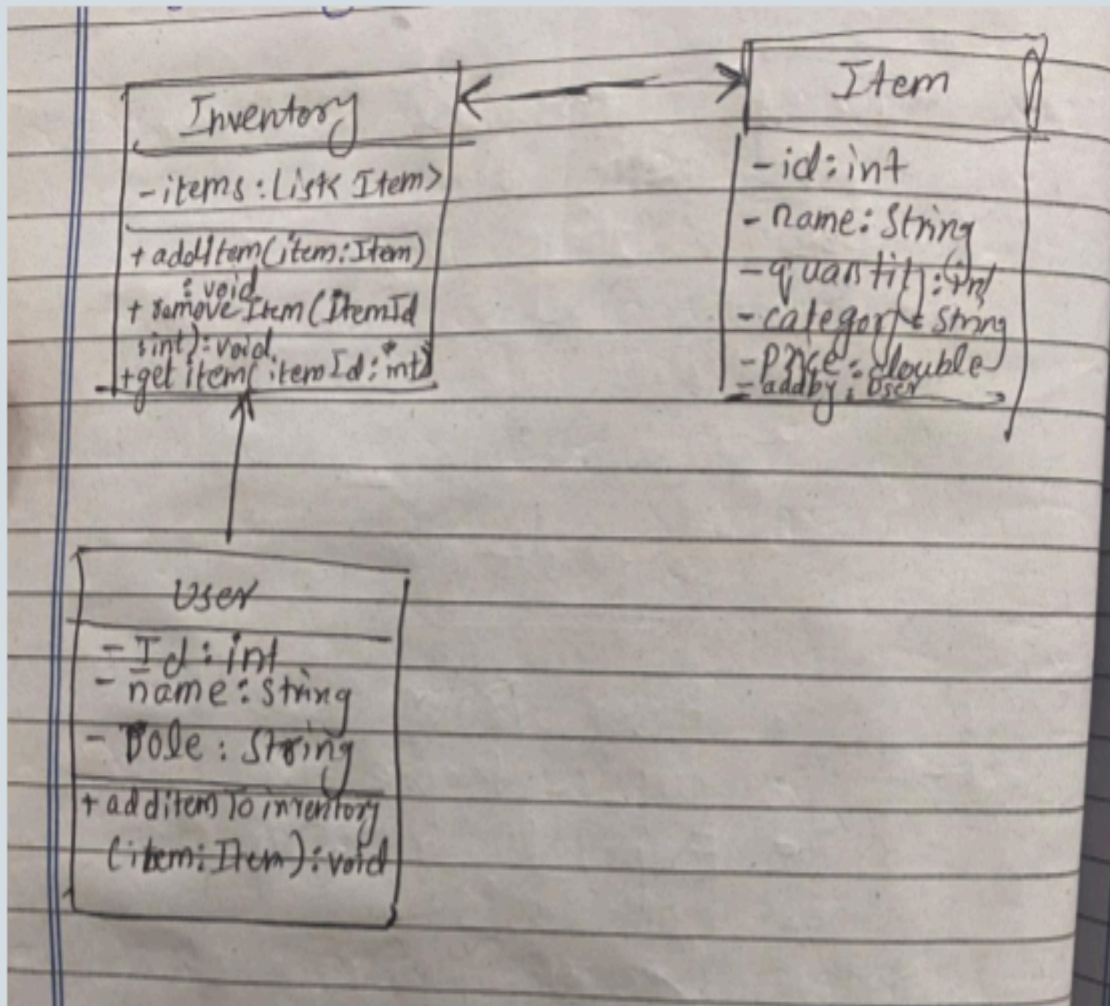
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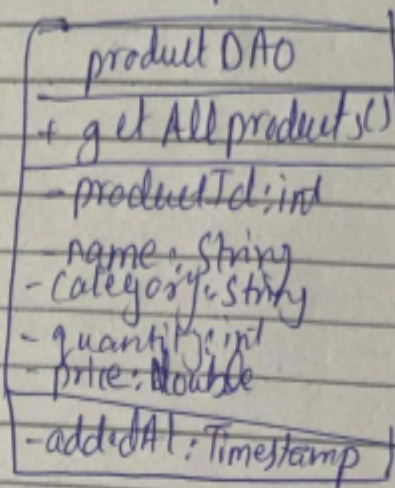
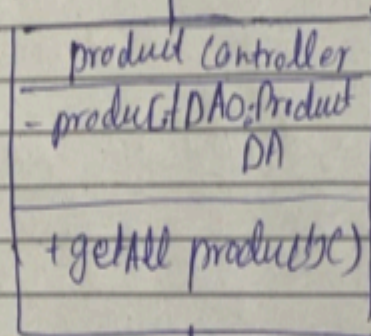
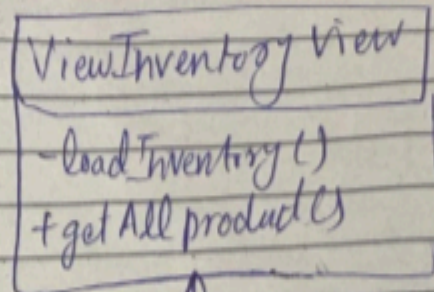








Class diagram



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